

SOWMYA P

Data Analyst | SQL • Python • Power BI • Machine Learning

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PROFESSIONAL SUMMARY

Data Analyst with a B.E. in Computer Science (Data Science) and hands-on experience across four end-to-end analytics and machine learning projects using Python, SQL, Power BI, and Streamlit to convert raw data into churn predictions, quantified KPIs, and stakeholder-ready dashboards. Skilled in exploratory data analysis (EDA), feature engineering, DAX, explainable AI (SHAP), and generative AI-assisted reporting, with a consistent track record of delivering accurate, decision-ready business insights.

CORE SKILLS & KEYWORDS

- **Languages & Querying:** Python (Pandas, NumPy, Matplotlib, Seaborn), SQL/MySQL (Joins, Window Functions, CTEs, Aggregations, Subqueries), Statistical Analysis
- **Machine Learning:** Scikit-learn, XGBoost, SHAP (Explainable AI), Feature Engineering, Streamlit Deployment
- **BI & Visualization:** Power BI (DAX, Power Query, Data Modeling, Dashboards, KPI Visuals, Drill-through), Microsoft Excel (Pivot Tables, VLOOKUP, Data Cleaning), Data Visualization
- **Data Analysis & BI:** Exploratory Data Analysis (EDA), Data Wrangling, Data Cleaning, Data Quality Assessment, KPI Reporting, Trend Analysis, Root-Cause Analysis, Data Storytelling, Dashboard Development, Data Modeling
- **AI & Automation:** Generative AI Tools (Groq API), Prompt Engineering, n8n Workflow Automation, API Integration
- **Tools:** Git, GitHub, Bitbucket, MySQL

WORK EXPERIENCE

AI Intern — Bharat Unnati AI Fellowship (Virtual) [GitHub](#)

Jan 2026 –

Apr 2026

- Engineered 3 AI-powered automation workflows for a restaurant business — review management, festival marketing, and food-caption generation — using n8n, Groq API, and JavaScript.
- Automated customer review handling by building sentiment analysis and escalation-detection logic, generating AI-driven responses synced to Google Sheets, and cut manual content-creation effort with a prompt-engineered marketing campaign generator.
- Identified task-completion efficiency gaps and informed workflow refinements by analyzing automation performance metrics in Excel.

PROJECTS

AI-Powered Customer Retention Intelligence Platform

Python • Pandas • Scikit-learn • XGBoost • SHAP • Streamlit • MySQL | [Live Demo](#)

- Built an end-to-end churn prediction platform on the 99K-customer Olist Brazilian e-commerce dataset, covering data cleaning, EDA, feature engineering, and machine learning.
- Engineered RFM and behavioral features (recency, frequency, monetary value, delivery and review behavior) to predict customer-level churn risk using Scikit-learn and XGBoost, and applied SHAP explainability to surface key risk drivers.
- Converted predictions into action by designing a rule-based retention recommendation engine and deploying a 4-module Streamlit app (Dashboard, Prediction, AI Insights, Executive Report).

E-Commerce Sales & Customer Analytics | [GitHub](#)

Python • Pandas • SQL • MySQL • Power BI • Excel

- Analyzed 99K+ e-commerce transactions across Python, SQL, and Power BI, identifying that São Paulo drove ~38% of 13.59M total revenue and credit cards accounted for 78% of payment volume.
- Uncovered the single largest revenue lever — a 3.12% repeat-customer rate against a 136.68 average order value — by building customer, product, and geographic KPI analysis in SQL and Power BI.
- Delivered a 4-page interactive Power BI dashboard (Executive Overview, Customer, Product, Business Insights) with quantified recommendations for loyalty programs and checkout optimization.

Credit Card Transaction & Customer Analytics | [GitHub](#)

Power BI • DAX • Power Query • Data Modeling

- Built a 2-report interactive Power BI dashboard analyzing credit card transaction performance and customer demographics using DAX measures and Power Query transformations.
- Enabled multi-dimensional business analysis by modeling revenue, interest earned, and transaction volume across card category, channel, income, education, and geography.

CERTIFICATIONS

- [Complete Python Bootcamp — Udemy](#)
- [Data Analytics Job Simulation — Forage](#)
- [Microsoft Office Masters Certification](#)

EDUCATION & ACHIEVEMENTS

B.E., Computer Science & Engineering (Data Science) — Sai Vidya Institute of Technology (2026) CGPA: 8.65

Pre-University (PCMC), Nagarjuna PU College — 80.5% (2022) | SSLC, Florence English High School — 79% (2020)

- Ranked 3rd in the 6th Semester of B.E. CSE (Data Science) based on academic performance.